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Article

HPV vaccine impact: genotype-specific changes in cervical pre-cancer share similarities with changes in cervical screening cytology

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Abstract

Background: After human papillomavirus (HPV) vaccine introduction, declines in the prevalence of HPV vaccine types have been observed in screening cytology, but data from the United States describing HPV type-specific changes in cervical intraepithelial neoplasia (CIN) grades 2-3 and adenocarcinoma in situ (CIN2/CIN3/AIS) are limited.

Methods: A statewide sample of individuals with cervical biopsies was selected for broad-spectrum HPV genotyping. CIN2/CIN3/AIS incidence and prevalence were calculated for type-specific high-risk HPV (hrHPV) among individuals aged 15-29 years. Weighted incidence rate ratios (IRR) and relative differences in prevalence (RDP) were computed to compare 3 time periods: 2006-2009 (Cohort 1 [C1], $n = 4121$), 2012-2015 (C2, $n = 2194$), and 2015-2018 (C3, $n = 1481$).

Results: When comparing C1 vs C3 among those aged 21-25 years, statistically significant reductions in hrHPV type-specific CIN2/CIN3/AIS incidence were observed for HPV16, HPV18, HPV31, and HPV33, with corresponding IRRs of 0.4 (95% confidence interval [95% CI] = 0.3 to 0.4), 0.3 (95% CI = 0.1 to 0.7), 0.6 (95% CI = 0.5 to 0.9), and 0.4 (95% CI = 0.1 to 0.8), respectively. The RDP comparing C1 vs C3 for HPV16/18-positive CIN2/CIN3/AIS was -43.8% ($P < .001$). When excluding HPV16/18 or HPV16/18/31/33 from all hrHPV types, the RDP was +56.6% and +92.5% ($P < .001$), respectively.

Conclusions: hrHPV type-specific CIN2/CIN3/AIS incidence decreased with statistical significance for vaccine types HPV16/18 and for HPV31 and HPV33. Although the HPV vaccine is highly beneficial and a top priority for preventing HPV-related cancer, the long-term vaccine impact in cohorts receiving the 4-valent HPV vaccine requires continued follow-up to assess genotype-specific distributions in the remaining CIN2+ lesions and cancers.

Introduction

Both randomized trials and population-based studies have shown the efficacy and effectiveness of human papillomavirus (HPV) vaccination in the reduction of pre-cancer defined as cervical intraepithelial neoplasia (CIN) grade 2 or grade 3 or worse (CIN2+, CIN3+).[1-6] In the United States, although HPV vaccine coverage has improved over the past 15 years,[7] it has been suboptimal, which has delayed and reduced the overall impact on cervical pre-cancer compared with that observed in other countries, including Sweden, England, Scotland, and Australia,[8-12] where higher HPV vaccine coverage was achieved early on. Estimated HPV vaccine coverage in New Mexico has increased over time with 53.1% of adolescent females receiving at least 1 dose in 2009, increasing to 66.7% in 2015 and to 82.0% in 2021.[13] US studies have nonetheless shown reductions in HPV-specific prevalence in cervical cytology and reductions in the incidence of CIN2+.[4,6,9,14-16]

Although monitoring overall trends in HPV prevalence, CIN incidence, and, ultimately, invasive cervical cancer is crucial, HPV type-specific data are essential for informing public health policy and to the design of next-generation cervical screening and triage approaches in the HPV vaccine era. Monitoring HPV type-specific prevalence and incidence in biopsy lesions is also important for evaluating vaccine cross-protection against non-vaccine HPV types, and to considering unmasking or type-

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replacement, particularly among the cohort of individuals receiving the 4-valent HPV vaccine (HPV 6/11/16/18).[17] Although there is evidence of vaccine impact in CIN from some countries, mainly in Europe,[18,19] data from the United States are limited.

Using real-world data from the state of New Mexico, where variable HPV vaccine uptake has been observed over time,[7] we examined HPV type-specific population-based incidence among individuals aged 15-29 years within 3 time period cohorts. In addition, we evaluated individual-level HPV vaccination and age at first dose for impact on CIN2+. This study extends findings from our previous work[20] by enabling a direct comparison of vaccine impact in cervical screening cytology vs cervical biopsies across a single large geographic area within the same US population.

Methods

Study population

Population-based cohorts of individuals with cervical biopsies were identified by the New Mexico HPV Pap Registry (NMHPVPR), a statewide public health surveillance program established in 2006 to assess all aspects of cervical cancer preventive care. Study cohorts included individuals with a cervix aged 15-29 years biopsied between January 2006 to December 2009 (Cohort 1; C1), January 2012 to June 2015 (Cohort 2; C2), and July 2015 to December 2018 (Cohort 3; C3). C1 represents an early or pre-HPV vaccine period, and C2 and C3 represent two post-HPV vaccine periods. Only biopsies of CIN2, CIN3, and adenocarcinoma in situ (AIS) were included in the analyses.

Formalin-fixed, paraffin-embedded (FFPE) cervical tissues were retrieved from individuals in each cohort (C1, C2 and C3) and subjected to broad-spectrum HPV genotyping (see Supplementary Methods). Biopsy specimens from laboratories performing pathology diagnoses on New Mexico residents were collected under research protocols approved by the University of New Mexico Human Research Review Committee.

Results were weighted to the total number of individuals with cervical biopsies in the population, stratified by biopsy diagnosis (CIN2 vs CIN3/AIS) and age (Table 1).

Laboratory methods

Human papillomavirus genotyping was performed on FFPE tissues as described previously[21,22] but with modifications as reported in Castro et al.[23] Both the Roche Linear Array (LA) HPV genotyping assay (Roche Diagnostics, Indianapolis, IN, USA) and the SPF10 PCR-DEIA-LiPA25, version 1 assay (Viroclinics-DDL [a Cerba Research company], the Netherlands) were used. The LA assay identifies 13 high-risk HPV (hrHPV) types (HPV16, 18, 31, 33, 35, 39, 45, 51, 52 [detectable only in the absence of HPV33, HPV35, or HPV58], 56, 58, 59, and 68) and 24 low-risk types (HPV6, 11, 26, 40, 42, 53-55, 61, 62, 64, 66, 67, 69, 70-73 [MM9], 81, 82 [MM4], 83 [MM7], 84 [MM8], IS39, and CP6108). The LiPA25, version 1 assay detects the same 13 hrHPV types but conglomerates the detection of HPV68 with HPV73 (HPV68/73). The LiPA25, version 1 also detects 12 low-risk types (HPV6, 11, 34, 40, 42, 43, 44, 53, 54, 66, 70, 74).

All biopsies negative for HPV using the LA assay were evaluated by Viroclinics-DDL for LiPA25, version 1 genotyping along with an interspersed random selection of HPV positive LA samples as controls. Investigators were blinded to both the LA results and associated tissue diagnoses. Positive genotyping results were combined from LA and LiPA25, version 1 assays for analyses as previously validated.[21]

Statistical methods

Our analysis focused on type-specific HPV incidence and prevalence comparisons between the 3 cohorts of individuals with endpoints of CIN2/CIN3/AIS and CIN3/AIS. When multiple hrHPV types were detected for an individual, attribution methods were applied (proportional or hierarchical; see Supplementary Methods).[24] For reference, the proportion of individuals with single hrHPV infections is detailed in Table S1.

Weighted hrHPV genotype-specific incidence rates and 95% confidence intervals (95% CI) per 100 000 individuals with screening cytology tests were calculated among individuals with CIN2/CIN3/AIS and CIN3/AIS. Incidence rate ratios (IRR) and corresponding 95% CIs were calculated comparing cohorts. Confidence intervals were constructed using 1000 bootstrap replications of the incidence rates to identify the distribution of the ratio, and intervals that did not include 1 were considered statistically significant. Weighted type-specific CIN2/CIN3/AIS incidence for HPV16, HPV18, HPV31, and HPV33 per 100 000 individuals with screening cytology tests, by year of biopsy, throughout all 3 time period cohorts (2006-2018), was plotted for those aged 21-25 years, and local regression lines fitted to show the trend.

Further, weighted absolute and relative differences in HPV type-specific prevalence, along with 95% CIs, were calculated. Taylor series expansions were used to estimate the variance of the relative weighted prevalence. *P* values were calculated to test if differences were statistically different from zero using the Z test for standard normal distributions. Differences were calculated between C1 and C2, and C1 and C3, for groups aged 15-20, 21-25, and 26-29 years with CIN2/CIN3/AIS and CIN3/AIS diagnoses. Differences were calculated for all individual hrHPV types, as well as groups (all hrHPV types, hrHPV types excluding HPV16/18, hrHPV types excluding HPV16/18/33, hrHPV types excluding HPV16/18/31/33 and HPV16/18).

Finally, logistic regression was used to estimate risk of developing CIN2+ for individuals who received at least 1 dose of the HPV vaccine compared with those with no vaccination. Estimates were calculated by age at first HPV vaccination. Trend tests were used to evaluate if age at vaccination was statistically significant.

All statistical tests were 2-sided, and *P* values less than .05 were considered statistically significant. Statistical analyses were conducted in SAS 9.4 (Cary, NC, USA), Stata 17.0 (College Station, TX, USA), and RStudio version 1.4.1717 (R Foundation for Statistical Computing).

Results

A large cohort of 7796 individuals (weighted) aged 15-29 years with CIN2/CIN3/AIS had HPV genotyping performed on tissue samples included in our analyses (Table 1). For reference, among those aged 21-25 years, there was no difference in the proportion of CIN2+ from all histological diagnoses (negative, CIN of any grade, or cancer) between the three time period cohorts ($P = .32$). Numbers of tissue samples with AIS diagnoses were small (Table 1).

Type-specific incidence rates for those with CIN2/CIN3/AIS and CIN3/AIS were computed, and proportional and hierarchical attribution was used for individuals with multiple hrHPV types detected. For those aged 21-25 years, the HPV16-related CIN2/CIN3/AIS incidence rate per 100 000 individuals with cervical screening tests was 681.9 (95% CI = 641.9 to 721.9) in C1, decreasing to 488.7 (95% CI = 438.4 to 538.9) in C2 and to 237.0 (95% CI =

Table 1. Sample and weighted numbers by age (15-20, 21-25, and 26-29 years) and histological diagnoses (CIN2 and CIN3/AIS) in the New Mexico population time period cohorts.

	Age (years)	Histology	Sample *N*	Sampling fraction and ratio (%)	Weighted *N*
Cohort 1 (C1)[a]	15-20	CIN2	449	65.2 (1.5)	689
January 2006 to December 2009		CIN3/AIS	179/1	66.2 (1.5)	270/2
		Total	629		961
	21-25	CIN2	786	64.0 (1.6)	1228
		CIN3/AIS	480/10	66.3 (1.5)	728/11
		Total	1276		1967
	26-29	CIN2	382	59.5 (1.7)	642
		CIN3/AIS	344/9	64.1 (1.6)	542/9
		Total	735		1193
	Total		2640		4121
Cohort 2 (C2)[b]	15-20	CIN2	39	57.4 (1.7)	68
January 2012 to June 2015		CIN3/AIS	15/0	68.2 (1.5)	22/0
		Total	54		90
	21-25	CIN2	311	51.8 (1.9)	601
		CIN3/AIS	278/3	55.4 (1.8)	499/7
		Total	592		1108
	26-29	CIN2	233	50.7 (2.0)	460
		CIN3/AIS	275/12	53.5 (1.9)	518/18
		Total	520		996
	Total		1166		2194
Cohort 3 (C3)[b]	15-20	CIN2	7	50.0 (2.0)	14
July 2015 to December 2018		CIN3/AIS	4/0	50.0 (2.0)	8/0
		Total	11		22
	21-25	CIN2	180	57.5 (1.7)	313
		CIN3/AIS	143/2	55.1 (1.8)	260/3
		Total	325		576
	26-29	CIN2	247	59.7 (1.7)	414
		CIN3/AIS	268/9	59.1 (1.7)	455/14
		Total	524		883
	Total		860		1481

[a] Cohort 1 (C1): 3829 individuals aged 15-29 years had HPV genotyping assays performed on biopsies with the highest diagnosis from 3832 colposcopy visits. Analysis excluded all biopsies with diagnoses of negative ($n = 222$), CIN1 ($n = 957$), and invasive cervical cancer ($n = 10$).
[b] Cohorts 2 (C2) and 3 (C3): 3629 individuals aged 15-29 years had HPV genotyping assays performed on biopsies with the highest diagnosis from 3629 colposcopy visits. Analyses excluded all biopsies with diagnoses of negative ($n = 3$), CIN1 ($n = 1572$), and ICC ($n = 28$). Biopsies were stratified by date to form two cohorts (ie, Cohort 2 (C2) spans January 2012 to June 2015 and Cohort 3 (C3) spans July 2015 to December 2018).
Abbreviations: AIS = adenocarcinoma in situ; CIN1 = cervical intraepithelial neoplasia [CIN] grade 1; CIN2 = CIN grade 2; CIN3 = CIN grade 3; HPV = human papillomavirus.

193.3 to 280.7) in C3 (Table 2). These rates corresponded to IRRs of 0.7 (95% CI = 0.6 to 0.8) comparing C1 and C2, and 0.4 (95% CI = 0.3 to 0.4) comparing C1 and C3. We observed greater reductions in incidence among the younger group aged 15-20 years, with HPV16-related CIN2/CIN3/AIS IRRs of 0.5 (95% CI = 0.3 to 0.7) when comparing C1 and C2, and an IRR of 0.1 (95% CI = 0.1 to 0.4) when comparing C1 and C3 (data not shown). Incidence rates increased among those in the older age group (26-29 years), with HPV16-related CIN2/CIN3/AIS incidence at 568.8 (95% CI = 526.1 to 611.5) per 100 000 individuals with cervical screening tests in C1, 624.1 (95% CI = 561.1 to 687.0) in C2, and 635.1 (95% CI = 563.8 to 706.4) in C3 (data not shown).

HPV type-specific incidence was plotted by year (2006-2018) for those aged 21-25 years with CIN2/CIN3/AIS (Figure 1). Large declines in HPV16-, HPV18-, HPV31-, and HPV33-related CIN2/CIN3/AIS incidence were observed. HPV16-related CIN2/CIN3/AIS incidence decreased from 951.1 per 100 000 individuals with cervical screening tests in 2006 to 566.4 in 2008, and then to 123.5 and 33.6 in 2012 and 2016, respectively. Although HPV18-related CIN2/CIN3/AIS incidence overall was much lower than that of HPV16, declines were still observed. The greatest decreases in incidence were seen when comparing CIN2/CIN3/AIS incidence rates for those aged 21-25 years between C1 and C3, in which the IRRs for HPV16, HPV18, HPV31, and HPV33 were 0.4, 0.3, 0.6, and 0.4, respectively (Table 2).

For those aged 21-25 years with CIN3/AIS, HPV16-related incidence decreased from 358.2 (95% CI = 334.1 to 382.2) cases per 100 000 individuals with cervical screening tests in C1 to 291.9 (95% CI = 256.6 to 327.2) in C2 and 147.9 (95% CI = 115.2 to 180.6) in C3, resulting in statistically significant IRR reductions (C1 vs C2: IRR = 0.8, 95% CI = 0.7 to 0.9, C1 vs C3: IRR = 0.4, 95% CI = 0.3 to 0.5) (Table 2). HPV18-related CIN3/AIS incidence rates were overall lower; in C1 the incidence rate was 11.4 (95% CI = 4.7 to 18.1) per 100 000 individuals with cervical screening tests and decreased to 8.7 (95% CI = 0.2 to 17.2) in C2 and 6.0 (95% CI = 0.0 to 14.1) in C3. Similar type-specific incidence patterns were seen among individuals with CIN3/AIS as in those with CIN2/CIN3/AIS but on a smaller scale.

Using proportional attribution for HPV type assignment among individuals with multiple co-infecting genotypes, the prevalence of HPV16 in those aged 21-25 years with CIN2/CIN3/AIS decreased from 45.2% in C1 to 36.8% in C2, and to 25.5% in C3 (Table 3). This corresponded to a relative reduction of 18.6% (95% CI = -25.8% to -11.4%, $P < .001$) between C1 and C2, and a relative reduction of 43.6% (95% CI = -51.8% to -35.4%, $P < .001$) between C1 and C3 (Table 3). The absolute differences were reductions of 8.4% (95% CI = -11.9% to -4.9%) between C1 and C2 and 19.7% (95% CI = -23.8% to -15.6%) between C1 and C3 (Table S2). Statistically significant prevalence reductions were also seen for HPV18 positive CIN2/CIN3/AIS, with greater reductions observed across time (relative difference for C1 vs C2: -41.6, 95% CI = -71.0% to -12.2%, $P = .01$ and for C1 vs C3: -47.2%, 95% CI = -83.3% to -11.1%, $P = .01$). Reductions were observed for HPV33, where the prevalence was 3.2% in C1, 3.3% in C2, and 2.2% in C3,

Table 2. HPV type-specific incidence rates and rate ratios (IRR) comparing population cohorts for individuals aged 21-25 years for groups of biopsy diagnoses (CIN2/CIN3/AIS and CIN3/AIS).

Biopsy diagnosis and HPV type	Cohort 1 (C1)[a] Incidence rate per 100 000 individuals with cervical screening tests (95% CI)	Cohort 2 (C2)[b] Incidence rate per 100 000 individuals with cervical screening tests (95% CI)	IRR (95%CI[d]) (Cohort 1 [C1] vs Cohort 2 [C2])	Cohort 3 (C3)[c] Incidence rate per 100 000 individuals with cervical screening tests (95% CI)	IRR (95% CI[d]) (Cohort 1 [C1] vs Cohort 3 [C3])
Biopsy: CIN2/CIN3/AIS					
HPV16	681.9 (641.9 to 721.9)	488.7 (438.4 to 538.9)	0.7 (0.6 to 0.8)	237.0 (193.3 to 280.7)	0.4 (0.3 to 0.4)
HPV18	47.0 (33.4 to 60.7)	22.2 (8.8 to 35.6)	0.5 (0.2 to 0.9)	14.8 (2.4 to 27.2)	0.3 (0.1 to 0.7)
HPV31	181.3 (156.1 to 206.6)	195.0 (158.4 to 231.6)	1.1 (0.9 to 1.3)	113.4 (81.2 to 145.6)	0.6 (0.5 to 0.9)
HPV33	49.9 (35.8 to 64.0)	41.0 (23.6 to 58.6)	0.8 (0.5 to 1.3)	20.3 (5.5 to 35.1)	0.4 (0.1 to 0.8)
HPV35	63.8 (47.8 to 79.8)	67.3 (45.0 to 89.6)	1.1 (0.7 to 1.7)	77.3 (51.0 to 103.7)	1.2 (0.8 to 1.9)
HPV39	57.1 (43.0 to 71.2)	47.6 (29.5 to 65.8)	0.8 (0.5 to 1.3)	55.6 (32.7 to 78.6)	1.0 (0.6 to 1.5)
HPV45	16.9 (8.5 to 25.3)	18.7 (6.98 to 30.6)	1.1 (0.4 to 2.5)	17.5 (4.0 to 30.9)	1.0 (0.3 to 2.5)
HPV51	84.1 (66.3 to 101.9)	54.9 (34.9 to 75.0)	0.7 (0.4 to 1.0)	55.8 (33.0 to 78.7)	0.7 (0.4 to 1.0)
HPV52	66.7 (51.0 to 82.4)	118.7 (89.5 to 147.8)	1.8 (1.3 to 2.5)	92.5 (62.9 to 122.2)	1.4 (0.9 to 2.0)
HPV56	15.3 (7.8 to 22.8)	20.4 (8.9 to 32.0)	1.3 (0.6 to 2.8)	15.4 (2.9 to 27.9)	1.0 (0.3 to 2.5)
HPV58	78.7 (61.3 to 96.1)	95.0 (68.4 to 121.7)	1.2 (0.8 to 1.8)	112.2 (80.0 to 144.4)	1.4 (1.0 to 2.1)
HPV59	21.2 (12.5 to 29.9)	27.2 (13.9 to 40.5)	1.3 (0.7 to 2.5)	30.6 (14.2 to 47.0)	1.4 (0.7 to 2.8)
HPV68	1.7 (0.0 to 4.2)	9.8 (1.6 to 18.0)	5.6 (1.2 to 55.8)	7.0 (0.0 to 15.2)	4.0 (0.7 to 40.8)
hrHPV	1365.7 (1340.1 to 1391.2)	1206.5 (1175.1 to 1237.9)	0.9 (0.9 to 0.9)	849.4 (821.4 to 877.4)	0.6 (0.6 to 0.7)
Biopsy: CIN3/AIS					
HPV16	358.2 (334.1 to 382.2)	291.9 (256.6 to 327.2)	0.8 (0.7 to 0.9)	147.9 (115.2 to 180.6)	0.4 (0.3 to 0.5)
HPV18	11.4 (4.7 to 18.1)	8.7 (0.2 to 17.2)	0.8 (0.2 to 2.0)	6.0 (0.0 to 14.1)	0.5 (0.0 to 1.8)
HPV31	75.8 (59.4 to 92.3)	94.1 (68.9 to 119.4)	1.2 (0.9 to 1.7)	55.1 (32.6 to 77.6)	0.7 (0.4 to 1.1)
HPV33	11.1 (4.5 to 17.7)	33.7 (18.2 to 49.3)	3.1 (1.5 to 7.0)	11.9 (0.5 to 23.3)	1.1 (0.2 to 3.3)
HPV35	15.7 (7.8 to 23.7)	29.0 (14.1 to 43.9)	1.9 (0.8 to 3.9)	28.8 (12.6 to 45.0)	1.8 (0.7 to 4.4)
HPV39	11.1 (4.7 to 17.4)	–[e]	–[e]	18.3 (4.4 to 32.2)	1.7 (0.5 to 4.3)
HPV45	6.6 (1.4 to 11.8)	12.8 (3.1 to 22.5)	1.9 (0.5 to 8.6)	3.4 (0.0 to 9.2)	0.5 (0.0 to 2.8)
HPV51	16.8 (8.6 to 24.9)	20.2 (8.4 to 32.0)	1.2 (0.5 to 2.5)	20.4 (6.3 to 34.4)	1.2 (0.5 to 2.7)
HPV52	11.2 (4.9 to 17.5)	38.3 (22.0 to 54.5)	3.4 (1.8 to 7.3)	39.3 (19.6 to 59.1)	3.5 (1.6 to 7.7)
HPV56	1.4 (0.0 to 3.7)	–[e]	–[e]	–[e]	–[e]
HPV58	20.6 (11.7 to 29.5)	36.3 (20.2 to 52.5)	1.8 (0.9 to 3.2)	57.6 (34.0 to 81.2)	2.8 (1.5 to 5.0)
HPV59	2.5 (0.0 to 5.8)	11.9 (3.1 to 20.7)	4.7 (1.1 to 113.3)	13.6 (2.0 to 25.2)	5.4 (1.1 to 123.2)
HPV68	–[e]	–[e]	–[e]	–[e]	–[e]
hrHPV	542.3 (530.9 to 553.7)	576.9 (560.8 to 593.1)	1.1 (1.0, 1.1)	402.3 (387.3, 417.2)	0.7 (0.7, 0.8)

[a] January 2006 to December 2009.
[b] January 2012 to June 2015.
[c] July 2015 to December 2018.
[d] Confidence intervals for weighted IRRs calculated using bootstrapped samples (1000 bootstrapped replications were produced to calculate the distribution of the rate ratio).
[e] No type-specific disease identified.
Data weighted to represent the statewide population of individuals with cervical biopsies. Number of individuals with screening cytology tests in the population cohorts were as follows: Cohort 1 (C1; $n = 129\,035$), Cohort 2 (C2; $n = 83\,186$), and Cohort 3 (C3; $n = 62\,226$).
HPV attribution was based on proportional attribution. Importance of hrHPV types was determined using the prevalence of hrHPV types in individuals with single hrHPV infections. Individual hrHPV type-specific prevalence was then multiplied for those with multiple infections. For example, when considering an individual with a CIN2 diagnosis and HPV16, 51, and 59 detected, the frequency of these hrHPV types in all individuals with CIN2 diagnoses was 12.1%, 14.0%, and 7.0%, respectively, summing to 33.1%. Using a proportional attribution approach, this case would attribute 36.6% to HPV16 (12.1%/33.1%), 42.3% to HPV51 (14.0%/33.1%), and 21.1% (7.0%/33.1%) to HPV59.
Abbreviations: AIS = adenocarcinoma in situ; CIN2 = cervical intraepithelial neoplasia [CIN] grade 2; CIN3 = CIN grade 3; HPV = human papillomavirus; hrHPV = high-risk HPV; IRR = incidence rate ratio; 95% CI = 95% confidence interval.

although the number of cases were small (63, 37, and 13, respectively). Combining HPV16 and HPV18 (HPV16/18), the relative reduction in HPV16/18 positive CIN2/CIN3/AIS was 43.8% (95% CI = -51.6% to -36.0%, $P < .001$) between C1 and C3, and yet, when excluding HPV16/18 from all other hrHPV types combined, a relative increase of 56.6% (95% CI = 44.2% to 69.1%, $P < .001$) was observed (Table 3). Nonsignificant changes were seen for all hrHPV types combined, which included HPV16/18/31/33 (relative difference +2.5%, $P = .09$). Statistically significant type-specific increases in CIN2/CIN3/AIS prevalence were observed for HPV35, HPV52, and HPV58 between C1 and C3 (all $P < .001$; Table 3). Prevalence differences were very similar when applying a hierarchical attribution for hrHPV types based on the 5-year positive predictive value for CIN3+ determined using data from a study of liquid-based cytology in the New Mexico population (Tables S3 and S4).[25]

Comparing individuals who received at least 1 dose of the HPV vaccine to those who did not, the risk of developing CIN2+ was significantly reduced if vaccinations were administered between the ages of 9 and 16 years (Table 4). Overall, there was an increasing risk of developing CIN2+ with increasing age at first HPV vaccination ($P_{trend} < .001$).

Discussion

Reductions in cervical pre-cancer incidence have been reported by work from the NMHPVPR[4,26] and others.[3,8,16,27] In particular, our results showed a 9.2% annual percent decrease in CIN2 incidence between 2007 and 2020 for individuals aged 20-24 years.[26] For completeness, Figure S1 shows the incidence of CIN2+ (CIN2/CIN3/AIS) between 2006 and 2018, where we observed a 46.1% overall cumulative reduction for individuals aged 21-25 years. These data correlate well with the population-based findings reported in our previous work, where a 49.4% reduction in the incidence of high-grade squamous intraepithelial lesions or worse (HSIL+) was

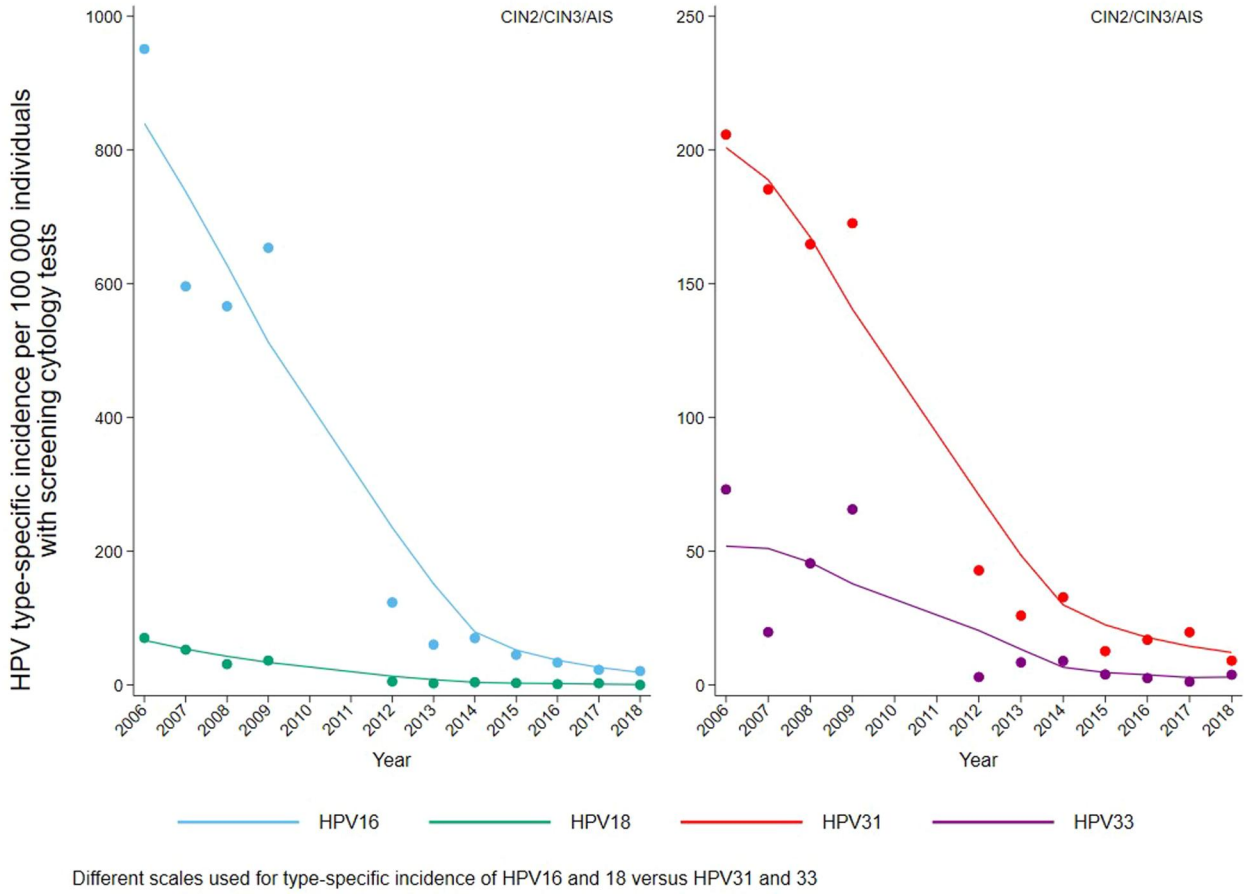


Figure 1. HPV type-specific incidence by year (2006-2018) among those aged 21-25 years with CIN2/CIN3/AIS. Data weighted to represent the statewide population of individuals with cervical biopsies. No data available for 2010/2011. Different scales used for type-specific incidence of HPV16 and 18 versus HPV31 and 33. Line graphs showing the trend in incidence of HPV type-specific infections between the years 2006 and 2018 for individuals aged 21-25 years with CIN2/CIN3/AIS excluding cancer. HPV types plotted were HPV16 and 18, and HPV31 and 33. Different scales were used for type-specific incidence of HPV16 and 18 versus HPV31 and 33. No data were available for the years 2010/2011 because they fell outside the cohorts for inclusion. Abbreviations: HPV = human papillomavirus; CIN2 = cervical intraepithelial neoplasia [CIN] grade 2; CIN3 = CIN grade 3; AIS = adenocarcinoma in situ.

observed in the source population's screening cytology from individuals aged 21-25 years across the same time period.[20]

For the group of individuals aged 21-25 years, we observed statistically significant reductions in CIN2/CIN3/AIS for HPV16 and HPV18. The IRR for HPV16 was 0.7 when comparing those aged 21-25 years biopsied in 2006-2009 (C1) to those biopsied in 2012-2015 (C2). This effect was augmented when comparing those biopsied in 2006-2009 (C1) to those biopsied in 2015-2018 (C3, IRR = 0.4), demonstrating that longer time since initial HPV vaccine implementation in the population is associated with greater increases in HPV vaccine type reductions. These data provide important information related to increasing the age of first cervical screening in the United States as cervical pre-cancer continues to decline significantly over time among the overall (vaccinated and unvaccinated combined) populations of younger individuals being screened. Furthermore, these data highlight how public health surveillance is critical to optimizing the integration of HPV vaccination and cervical cancer screening in the absence of accurate individual-level HPV vaccination information, a continuing reality across the United States.

HPV16/18-related CIN2/CIN3/AIS reductions among those aged 20-24 years have been reported by the HPV-Vaccine Impact Monitoring Project (HPV-IMPACT) project, but overall vaccine impact was not reported when excluding HPV genotypes with observed type-specific reductions (eg, HPV16/18/31).[14] We observed an increase in prevalence when excluding HPV16/18 and HPV16/18/31/33 from all hrHPV types (relative difference +56.6% and +92.5%, respectively, $P < .001$; Table 2), with non-significant changes in prevalence observed for all hrHPV types (relative difference +2.5%, $P = .09$). For older individuals, our study and the HPV-IMPACT project[14] found increases in HPV type-specific CIN2/CIN3/AIS incidence rates, which could be due to guideline changes recommended across our surveillance periods[28,29] or changes in sexual behaviors between age-specific cohorts.[30] Guidelines updated screening intervals, clinical screening approaches, and clinical management algorithms where individuals with normal cytology but HPV16/18/45 positive genotyping results are sent directly to colposcopy.

Comparing hrHPV type-specific incidence of CIN2/CIN3/AIS between 2006-2009 (C1) and 2015-2018 (C3), we observed IRRs of 0.6 and 0.4 for HPV31 and HPV33, respectively, supporting HPV vaccine cross-protection also observed in the screening cytology studies from this population reported in our previous work.[20] The HPV-IMPACT project also showed reductions in CIN2/CIN3/AIS IRRs of 0.5 and 0.6 for HPV31 and 33, respectively.[14] These population-based data support observations from randomized

Table 3. HPV type-specific and group prevalence and relative prevalence difference comparing population cohorts for individuals aged 21-25 years with CIN2/CIN3/AIS diagnoses.

HPV type	Cohort 1 (C1)[a] ($n = 1967$)		Cohort 2 (C2)[b] ($n = 1108$)		Relative prevalence difference (95% CI)	P[d]	Cohort 3 (C3)[c] ($n = 576$)		Relative prevalence difference (95% CI)	P[d]
	N	Prevalence (%)	N	Prevalence (%)			N	Prevalence (%)		
HPV16	894	45.2	412	36.8	−18.6 (−25.8 to -11.4)	<.001	146	25.5	−43.6 (−51.8 to -35.4)	<.001
HPV18	60	3.1	20	1.8	−41.6 (−71.0 to -12.2)	.01	9	1.6	−47.2 (−83.3 to -11.1)	.01
HPV31	237	12.1	163	14.7	+21.9 (−0.7 to +44.5)	.06	71	12.4	+2.7 (−22.7 to +28.2)	.83
HPV33	63	3.2	37	3.3	+2.4 (−38.2 to +43.0)	.91	13	2.2	−31.6 (−72.4 to +9.3)	.13
HPV35	81	4.2	56	5.1	+22.7 (−18.0 to +63.3)	.27	49	8.5	+104.8 (+34.9 to +174.6)	<.001
HPV39	71	3.6	35	3.2	−10.6 (−45.9 to +24.7)	.56	33	5.8	+59.0 (−4.6 to +122.6)	.07
HPV45	22	1.1	17	1.5	+32.8 (−51.0 to +116.5)	.44	11	1.9	+69.4 (−51.5 to +190.4)	.26
HPV51	106	5.4	45	4.1	−24.2 (−49.9 to +1.5)	.06	36	6.2	+14.5 (−27.4 to +56.3)	.50
HPV52	81	4.2	96	8.8	+110.5 (+50.6 to +170.5)	<.001	58	10.0	+140.4 (+62.5 to +218.2)	<.001
HPV56	19	1.0	16	1.5	+48.5 (−49.8 to +146.8)	.33	11	1.9	+90.1 (−49.3 to +229.4)	.21
HPV58	100	5.1	77	7.0	+37.3 (−2.3 to +76.8)	.06	69	12.0	+135.5 (+66.6 to +204.5)	<.001
HPV59	26	1.4	23	2.1	+52.8 (−32.0 to +137.5)	.22	18	3.1	+130.0 (−6.2 to +266.2)	.06
HPV68	2	0.1	7	0.6	+481.1 (−412.8 to +1375.0)	.29	4	0.8	590.8 (−536.7 to +1718.4)	.30
hrHPV	1764	89.6	1004	90.5	+1.0 (−1.5 to +3.4)	.45	529	91.8	+2.5 (−0.4 to +5.4)	.09
hrHPV excl. 16/18	810	41.3	572	51.9	+25.5 (+15.9 to +35.1)	<.001	373	64.7	+56.6 (+44.2 to +69.1)	<.001
hrHPV excl. 16/18/33	747	38.1	535	48.6	+27.5 (+17.0 to +37.9)	<.001	361	62.5	+64.1 (+50.3 to +77.9)	<.001
hrHPV excl. 16/18/31/33	509	26.0	372	33.9	+30.0 (+15.8 to +44.3)	<.001	289	50.1	+92.5 (+71.5 to +113.5)	<.001
HPV16/18	954	48.3	431	38.6	−20.1 (−26.9 to -13.3)	<.001	155	27.1	−43.8 (−51.6 to -36.0)	<.001

[a] January 2006 to December 2009.
[b] January 2012 to June 2015.
[c] July 2015 to December 2018.
[d] *P* values are 2-sided and based on the standard normal Z-test.

Data weighted to represent the statewide population of individuals with cervical biopsies.

HPV attribution was based on proportional attribution. Importance of hrHPV types was determined using the prevalence of hrHPV types in individuals with single hrHPV infections. Individual hrHPV type-specific prevalence was then multiplied for those with multiple infections. For example, when considering an individual with a CIN2 diagnosis and HPV16, 51, and 59 detected, the frequency of these hrHPV types in all individuals with CIN2 diagnoses was 12.1%, 14.0%, and 7.0%, respectively, summing to 33.1%. Using a proportional attribution approach, this case would attribute 36.6% to HPV16 (12.1%/33.1%), 42.3% to HPV51 (14.0%/33.1%), and 21.1% (7.0%/33.1%) to HPV59.

Abbreviations: AIS = adenocarcinoma in situ; CIN2 = cervical intraepithelial neoplasia [CIN] grade 2; CIN3 = CIN grade 3; HPV = human papillomavirus; hrHPV = high-risk HPV; 95% CI = 95% confidence interval.

Table 4. Odds of a diagnosis of cervical intraepithelial neoplasia grade 2 (CIN2+) by human papillomavirus (HPV) vaccination status and age at first vaccination.

Vaccine status	Age at vaccination (years)	N screened	N (%) CIN2+	OR[a] (95% CI)	P[b]
Unvaccinated	–	80 322	2101 (2.62)	*Ref*	
Vaccinated	9-12	5881	58 (0.99)	0.37 (0.28 to 0.48)	<.001
	13-14	8011	89 (1.11)	0.38 (0.31 to 0.47)	<.001
	15-16	9029	144 (1.59)	0.55 (0.46 to 0.65)	<.001
	17-18	6539	182 (2.78)	1.06 (0.91 to 1.24)	.438
	≥19	2318	72 (3.11)	1.84 (1.45 to 2.35)	<.001

[a] Adjusted for age at screening.
[b] P values are two sided and based on the standard normal Z-test.
Data restricted to individuals born in the birth cohort 1989-1996 in alignment with other study analyses. Only individuals with vaccination records prior to age 25 years and prior to any cytology or biopsy reports submitted to the New Mexico HPV Pap Registry were considered vaccinated. CIN2+ outcomes were the highest biopsy record ever reported for an individual.
Abbreviations: CIN2+ = cervical intraepithelial neoplasia [CIN] grade 2 or worse; CI = confidence interval; OR = odds ratio.

trials and clinical studies.[17,31] For example, in the Females United to Unilaterally Reduce Endo/Ectocervical Disease (FUTURE) trial, reductions of 33.6% for HPV31 and 22.5% for HPV33 were observed when comparing those vaccinated to those unvaccinated.[17] Clinical trials have also found evidence of cross-protection against other hrHPV types, including HPV35, HPV45, and HPV51, with differences dependent on the HPV vaccine under evaluation.[32] Although not statistically significant, we did observe decreases in CIN2/CIN3/AIS incidence for HPV39 and HPV51.

Our findings from a large population within a single geographic area contribute to considerations of HPV type-replacement or unmasking, 2 distinct competing disease risks discussed previously when reporting the results of a phase III clinical trial of the 4-valent HPV vaccine.[17] HPV type-replacement within a biologic niche (eg, cervix, vaginal pool) would be assigned when alterations in the overall fitness or carcinogenicity of nonvaccine HPV types would occur. This is not supported by evolutionary data. Bacterial agents with large genomes and viruses with high mutation rates can readily drive genetic adaptations to improve fitness; however, HPVs demonstrate little capacity for mutation given evolutionary saturation.[33]

In contrast to HPV type-replacement, unmasking of disease risk refers to when co-infections of related etiological agents are initially unassigned due to sampling error, detection bias, or, in this instance, potentially from removal of viral exclusion pressures.[17,34-37] We previously suggested that nonvaccine HPV type-specific co-infections could be anticipated as competing disease risks in vaccinated populations through unmasking.[17] Increases in the detection of other hrHPV infections in cervical screening samples[20] and CIN2/CIN3/AIS are important to populations receiving the 4-valent HPV vaccine in the United States and elsewhere, as well as in populations receiving the bivalent HPV vaccine.[2,11] In the long-term follow-up of the Costa Rica vaccine study, higher rates of CIN2+ events and CIN3+ events due to nonpreventable HPV types in vaccinated vs unvaccinated participants suggested clinical unmasking.[38] Although the overall benefit of HPV vaccine is high, unmasking may reduce the overall anticipated impact on CIN2/CIN3/AIS and invasive cervical cancer. This will be important to continue to examine. Subsequent implementation of the 9-valent vaccine, with broader HPV coverage, is expected to remove or mute these potential concerns. We do note, however, that HPV type-specific prevalence varies by geographic location worldwide and some populations may not benefit as greatly from the 9-valent HPV vaccine; for example, HPV35 is not included in the 9-valent vaccine, yet more prevalent in sub-Saharan Africa.[39]

Although worldwide the vast majority of invasive cervical cancer is attributed to HPV16 and 18 infections, it is relevant here to mention that non-HPV16/18 genotypes in cervical cancers have been associated with older age at diagnosis compared with HPV16/18 cancers,[21,40] likely due to lower carcinogenicity of these genotypes. As a result, in the United States, the full impact of unmasking of cancers caused by nonvaccine types may be delayed with a slight shift to older ages in the peak of cervical cancer incidence among the decade of individuals receiving the 4-valent HPV vaccine. Although few study settings in the United States can evaluate the HPV type-specific impact of vaccination on HPV prevalence in CIN, and even fewer can also measure the HPV type-specific impact of vaccination in screening cytology among the same population, HPV type-specific effects on invasive cervical cancer will be more difficult to observe due to overall low population incidence and the long sojourn time from HPV infection to cancer diagnoses.[40,41] Reductions in overall incidence rates of invasive cervical cancer have recently been reported in younger individuals aged 15-20 years[15] and 20-24 years,[42] age groups no longer recommended for cervical screening by the American Cancer Society.[28]

One limitation of our study comes from a potential biopsy referral bias of HPV16/18-positive individuals. We have not directly correlated the clinical HPV tests used for those in our cohorts, yet HPV16/18 positive individuals should be referred directly to colposcopy under current US management guidelines.[28,29] However, HPV genotyping was not routinely nor commonly used in individuals less than 30 years of age in our population during the period of this current investigation.[43]

To conclude, we found statistically significant reductions in the incidence of CIN2/CIN3/AIS for vaccine types HPV16 and 18, as well as for HPV31 and 33, supporting expectations of large reductions in HPV-related cancers. Uniquely, this investigation enables direct comparison of population-based HPV type-specific data generated from biopsy lesions vs screening cytology specimens in the same statewide population.[20]

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Author contributions

Rachael Adcock (Data curation, Formal analysis, Writing—original draft, Writing—review & editing), Cosette M. Wheeler (Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing—original draft, Writing—review & editing), William Curtis Hunt (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing—review & editing), Norah Torrez-Martinez (Data curation, Investigation, Writing—review & editing), Michael Robertson (Data curation, Investigation, Methodology, Writing—review & editing), Ruth McDonald (Data curation, Writing—review & editing), Nancy E. Joste (Conceptualization, Investigation, Methodology, Writing—review & editing), Mark H. Stoler (Conceptualization, Investigation, Methodology, Writing—review & editing), Maurits N.C. de Koning (Investigation, Methodology, Writing—review & editing), and Wim G.V. Quint (Investigation, Methodology, Writing—review & editing)

Supplementary material

Supplementary material is available at *JNCI: Journal of the National Cancer Institute* online.

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Conflicts of interest

C.M.W. and J.C. have received funds from grants, cooperative agreements, or subcontracts related to cervical screening and triage through their respective institution, the University of New Mexico Health Sciences Center and Queen Mary University London. C.M.W. reports receiving reagents and equipment from Roche Molecular Systems, Roche/Ventana Medical Systems, and Hologic through her institution and outside of the submitted work, and research support from Hologic and Becton Dickinson (BD), also outside of this work. J.C. reports being on the speaker's bureau for Hologic and BD and the data monitoring committee for a Roche liver cancer trial. M.K. has received grants from Viroclinics-DDL (a Cerba Research company) outside the work of the current study. M.H.S. is a consultant in clinical trial development and performance and/or an expert pathologist in clinical trials for Merck, Roche/Ventana Medical Systems, Becton Dickinson (BD) Life Sciences, Abbott Molecular, Inovio Pharmaceuticals, and Frantz Viral Therapeutics. All other authors and NMHPVPR Steering Committee members have nothing to disclose.

Data availability

Data devoid of personal information can be requested through contacting the University of New Mexico Health Sciences Center Pre-Award Office. Execution of a data use agreement (DUA) is required.

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